

Car Rental System - Database Requirements

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Non-Functional Requirements

ID	Requirement (MoSCoW)	Description	Justification
NFR-1	Performance – MUST Have	The database must handle at least 500 concurrent user queries with minimal delay (response time < 2 seconds).	Ensures quick access for customers and staff during peak booking times.
NFR-2	Availability – MUST Have	The system must maintain 99.9% uptime, ensuring that the service is accessible at all times.	The car rental business operates continuously and needs constant availability to handle reservations.
NFR-3	Security – MUST Have	Customer data (e.g., driver's license, payment info) must be encrypted and access-controlled.	Protects sensitive user data and ensures compliance with data protection regulations (e.g., GDPR).
NFR-4	Scalability – SHOULD Have	The database should support future expansion, allowing addition of new branches, cars, and users without major reconfiguration.	Ensures smooth business growth and flexibility.
NFR-5	Backup & Recovery – MUST Have	The system must perform automated daily backups and allow recovery within one hour in	Prevents data loss due to hardware failure or human error, maintaining

		case of data loss.	business continuity.
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Functional Requirements

ID	MoSCoW Category	Functional Requirement	Justification
FR-1	MUST Have	Register new customers	The system must allow registration of new users with personal and license information.
FR-2	MUST Have	Authenticate users	The system must allow users to securely log in using their credentials.
FR-3	MUST Have	Search available cars	The system must allow users to search cars by model, price, and availability.
FR-4	MUST Have	Book a car	Users must be able to book cars for specific dates and times.
FR-5	MUST Have / ASK Should Have	Manage bookings	Customers can view, modify, or cancel their bookings.
FR-6	MUST Have	Process payments	The system must process credit card or cash payments securely.
FR-7	MUST Have	Generate invoices	The system must automatically generate invoices for completed rentals.
FR-8	MUST Have	Track rental status	Staff should be able to track which cars

			are rented, available, or under maintenance.
FR-9	SHOULD Have	Support discounts	The system should allow promotional codes or discounts.
FR-10	MUST Have	Record car details	The database must store detailed information for each car including model, license plate, and mileage.
FR-11	MUST Have	Record customer feedback	The system must allow customers to leave feedback about their rental experience.
FR-12	COULD Have	Send notifications	The system could notify users via email or SMS about bookings or due dates.
FR-13	MUST Have	Generate reports	The system must generate reports on revenue, usage, and availability.
FR-14	MUST Have	Manage staff accounts	Admins must be able to create and manage employee accounts.
FR-15	SHOULD Have	Track maintenance schedules	The database should track each car's maintenance history and next service date.
FR-16	MUST Have	Handle returns	The system must record when cars

			are returned and update their availability.
FR-17	SHOULD Have	Calculate rental cost	The system should automatically calculate total rental costs including taxes and insurance.
FR-18	COULD Have	Store branch information	The database could store details about each branch location and contact info.
FR-19	MUST Have	Log system activity	The system must log all database updates and user actions for audit purposes.
FR-20	SHOULD Have	Integrate with insurance provider	The system should connect to third-party APIs to verify car insurance coverage.